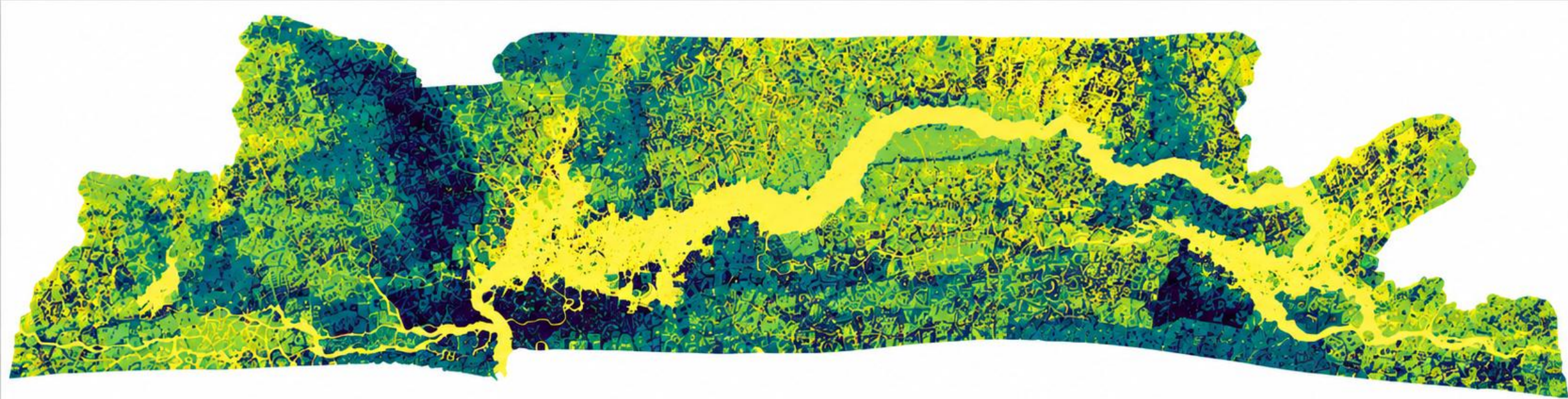




# Interpretable Multi-Source Machine Learning for **Urban Classification** in Data-Scarce Regions

Peter Edemewe Ugege • John Aigbokhan • Emmanuel Akalumhe

Forestry Research Institute of Nigeria (FRIN)

Deep Learning Indaba 2026

Research Track | Oral Presentation



DEEP  
LEARNING  
INDABA  
2026



# Why Is Accurate Urban Mapping Still So **Difficult**?



## Rapid Urban Expansion

Rapid urban growth outpaces mapping capacity.



## Highly Heterogeneous Landscapes

Mixture of dense built-up, informal settlements, vegetation, water, and bare surfaces.



## Limited High-Quality Training Data

Scarcity of labeled data and inconsistent ground truth reduce model reliability.



## Black-Box Models Are Difficult to Trust

Black-box models lack transparency and actionable insights.



## Poor Generalization Across Regions

Models trained in one area often fail to generalize to new, unseen environments.



**Our Solution: Interpretable Multi-Source Machine Learning.**

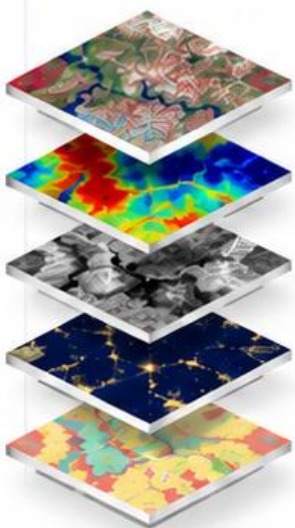


# Our Approach

An interpretable, multi-source machine learning framework for reliable urban classification in data-scarce regions.

1

## Multi-Source Data Integration



- Sentinel-2 (Optical)
- LST (Thermal)
- DEM & Topography
- Built-up & Open Data
- Ancillary Data



2

## Feature Engineering

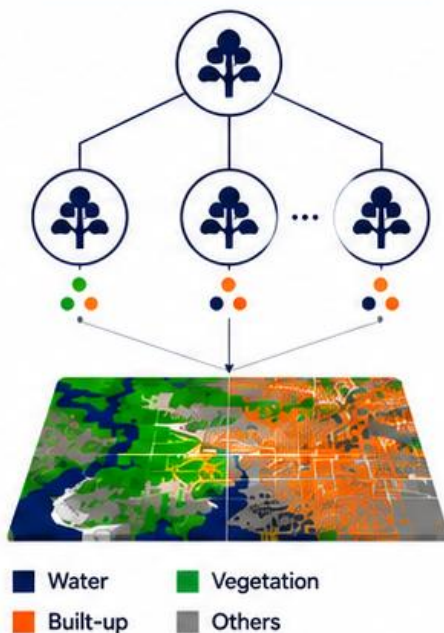


- Spectral Indices
- Texture Metrics
- Morphological Features
- Environmental Variables



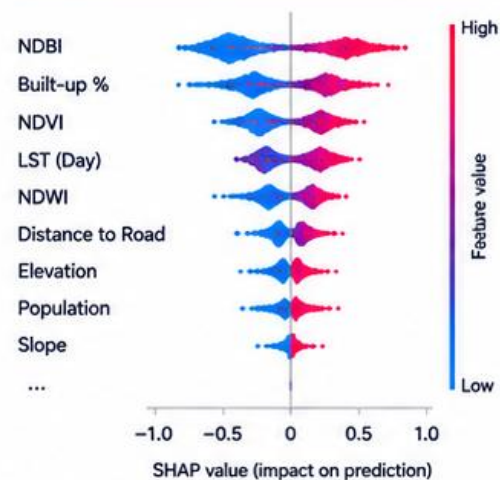
3

## Random Forest Classification



4

## Explainable AI (SHAP)



Transparent insights into what drives each prediction.



5

## Reliable Urban Classification



- ✓ High Accuracy
- ✓ Interpretable Predictions
- ✓ Generalizable Across Regions
- ✓ Supports Data-Driven Decisions



**Our Goal:** Accurate, explainable, and transferable urban mapping using diverse data and interpretable AI.

# Multi-Source Feature Integration

*Eleven complementary predictors were integrated into a unified feature space.*



**Study Area: Lagos State, Nigeria**



**Feature Groups**



**Spectral Features**

- Blue
- Green
- Red
- NIR
- SWIR
- NDVI



**Urban Context**

- NDBI
- Population
- Nightlights



**Physical Context**

- Elevation



**Temporal Context**

- Year



**Feature Matrix**

*11 predictor variables per sampled location*



**Random Forest Classifier**

*Urban / Non-Urban Prediction*



Multi-source feature integration improved **accuracy** and **interpretability**.

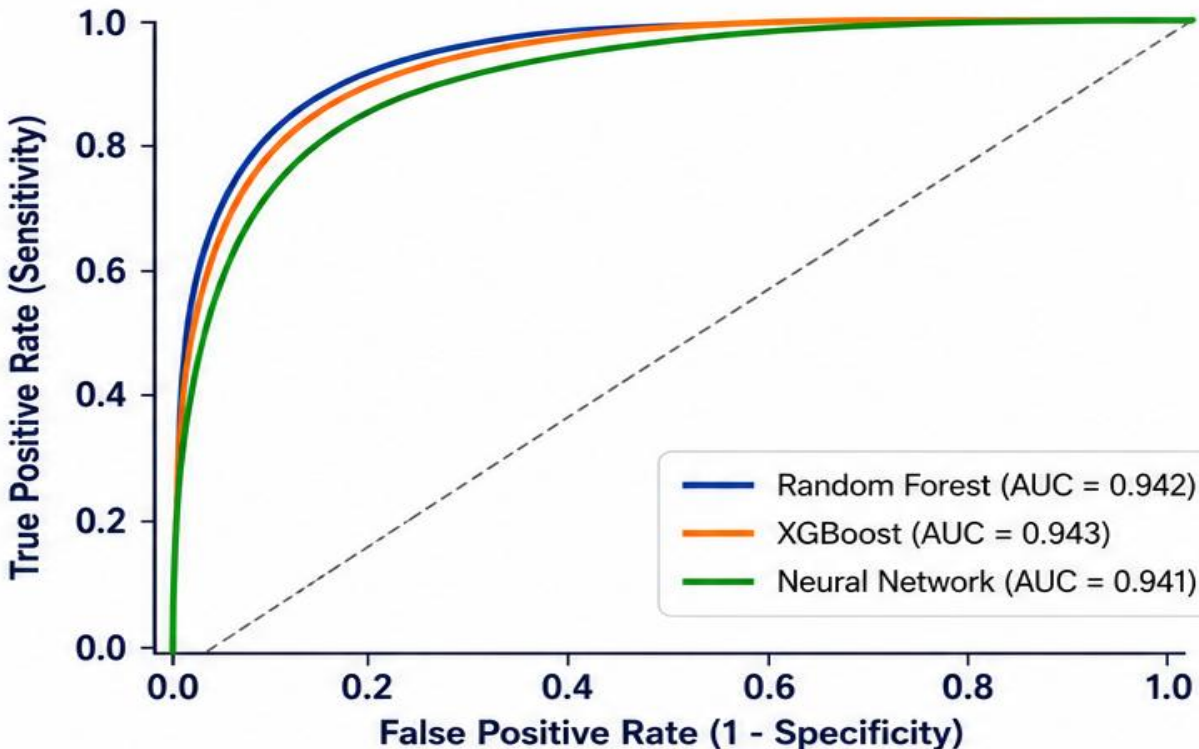


# Model Performance Evaluation

Three machine learning models demonstrated consistently strong classification performance.



ROC Curve Comparison for Urban Classification



ROC curves demonstrate nearly identical discrimination performance across the three models.



Performance Summary

| Model                                                                                               | Accuracy | Precision | Recall | AUC    |
|-----------------------------------------------------------------------------------------------------|----------|-----------|--------|--------|
|  Random Forest   | 0.87     | 0.87      | 0.87   | 0.942  |
|  XGBoost         | 0.87     | 0.87      | 0.87   | 0.943★ |
|  Neural Network | 0.86     | 0.87      | 0.86   | 0.941  |

★ Best performance



All three models achieved consistently high classification performance, with **AUC values above 0.94**.

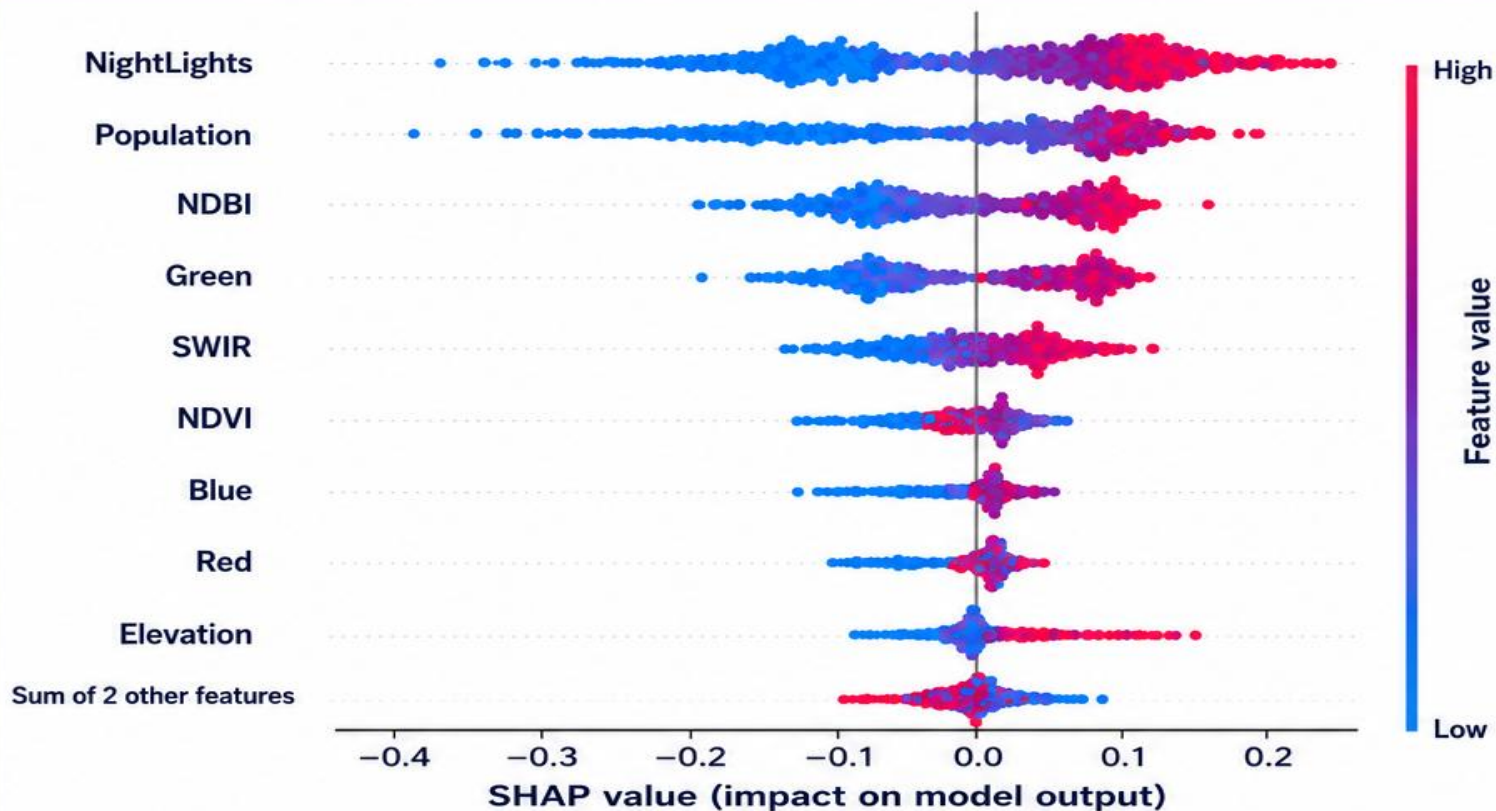


# Model Explainability with SHAP

SHAP reveals both the importance of each predictor and its influence on urban classification decisions.



## SHAP Summary Plot (Impact on Model Output)



Each dot represents a sample. Color shows the feature value (blue = low, red = high). Position shows the impact on prediction (right = increases, left = decreases).



## Key Insights



### Nightlights

Most influential predictor; high values strongly increase urban probability.



### Population

High population consistently drives predictions toward urban areas.



### NDBI & Spectral Bands

Provide critical information that refines urban boundaries and built-up patterns.

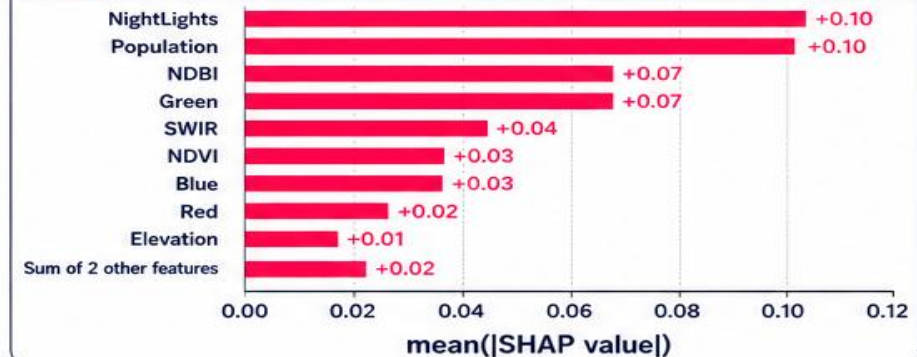


### Elevation

Least influential; minimal effect on model decisions in this low-relief area.



## Feature Importance Ranking (mean |SHAP value|)



**Urban activity indicators** (Nightlights, Population, NDBI) are the dominant drivers of classification, while **spectral and terrain variables** provide complementary information.



# Key Findings

Three key insights from the proposed framework



1

## High Predictive Performance

All three machine learning models achieved consistently strong classification performance (AUC > 0.94).



2

## Explainable AI Revealed Meaningful Urban Drivers

Nightlights, Population and NDBI emerged as the dominant drivers of urban classification.



3

## Practical Framework for Data-Scarce Regions

The framework is **transferable** to data-scarce regions using openly available geospatial data.

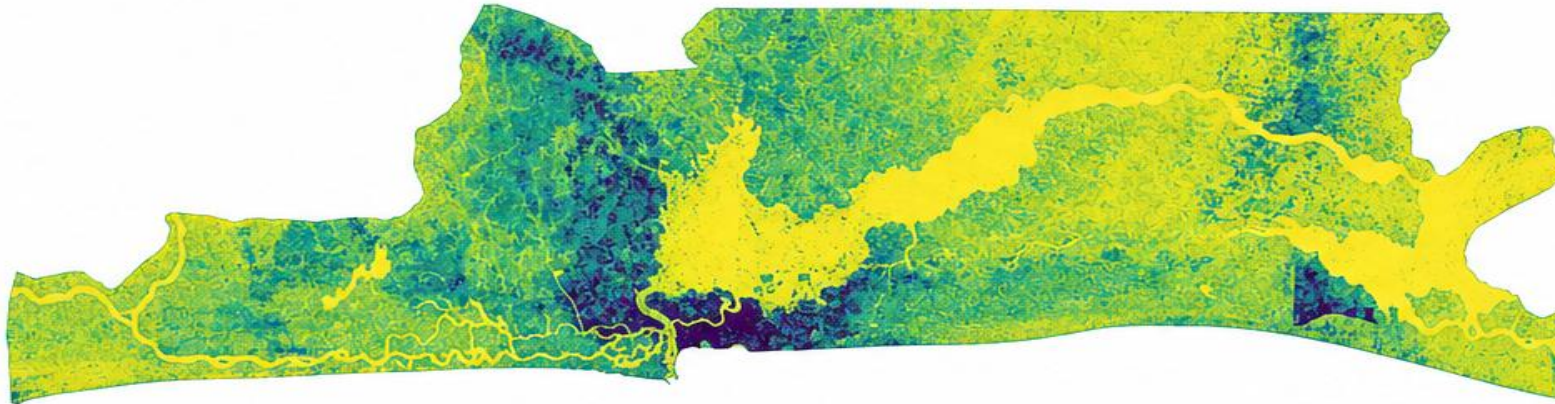


Multi-source machine learning delivers **accurate, interpretable** and **actionable** urban mapping in data-scarce environments.



# From Prediction to Decision Support

*Accurate, interpretable urban intelligence for data-scarce regions*



**Urban Probability**

Low      Moderate      High      Very High



Map shows predicted probability of being urban in Lagos, Nigeria (2020) derived from interpretable multi-source machine learning.

## Why it matters



### Accurate Urban Monitoring

High-resolution, cost-effective mapping enables up-to-date urban monitoring.



### Explainable Decision Support

Interpretable AI reveals key urban drivers, building trust for informed decisions.



### Transferable Framework

Developed using openly available data, applicable to other data-scarce regions.



### Supports Sustainable Planning

Provides evidence for resilient, inclusive and sustainable urban development.



Interpretable multi-source machine learning enables **accurate, transparent** and **scalable** urban mapping for data-scarce regions.





# Thank You!

## *Questions & Discussion*



**Peter Ugege**

Informatics & ICT Resource Centre  
Forestry Research Institute of Nigeria (FRIN)



pugege@frin.gov.ng



ORCID: 0000-0002-2722-0076

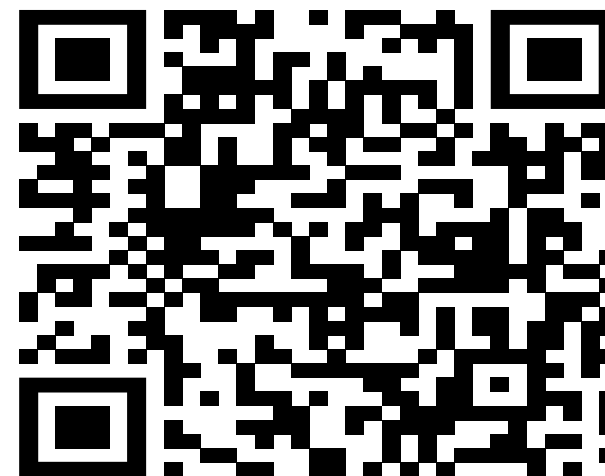


Ibadan, Nigeria



**Scan to Read the Paper**

Scan the QR code to read  
the full paper.



**DLI** DEEP LEARNING  
**INDABA 2026**  
*The African Deep Learning Conference*



2 – 7 August 2026



Pan-Atlantic University  
Lagos, Nigeria